Exponential Functions and Their Graphs

ANSWER KEY

Evaluating and describing

1 Let $f(x) = 3 \cdot 2^x$. Evaluate:

a $f(0) \;\; = 3$ **b** $f(3) \;\; = 24$ **c** $f(-2) \;\; = \frac{3}{4}$

2 For $y = 5^x$, state:

a the domain All real numbers.

b the range $y > 0$.

c the horizontal asymptote $y = 0$.

d the y-intercept $(0, 1)$.

3 Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of g.

Shift right 3 units and up 1 unit. Horizontal asymptote: $y = 1$.

Identifying from graphs and applications

4 The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.

$y = 2^x$.

5 An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%: $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest riyal.

$A(5) = 1000e^{0.3} \approx 1350$ riyals.

6 Determine whether each function represents growth or decay:

a $y = 4(1.08)^x$ Growth.

b $y = 7(0.85)^x$ Decay.

c $y = 2e^{-0.3x}$ Decay.

d $y = \left(\frac{5}{4}\right)^x$ Growth.